

King's Digital Lab

Creating digital tools to explore academic research in new ways.

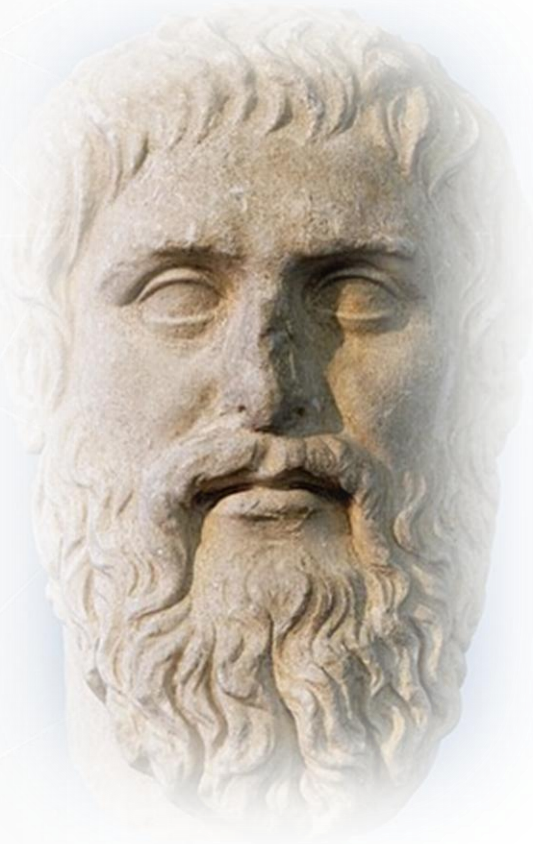
kdl.kcl.ac.uk



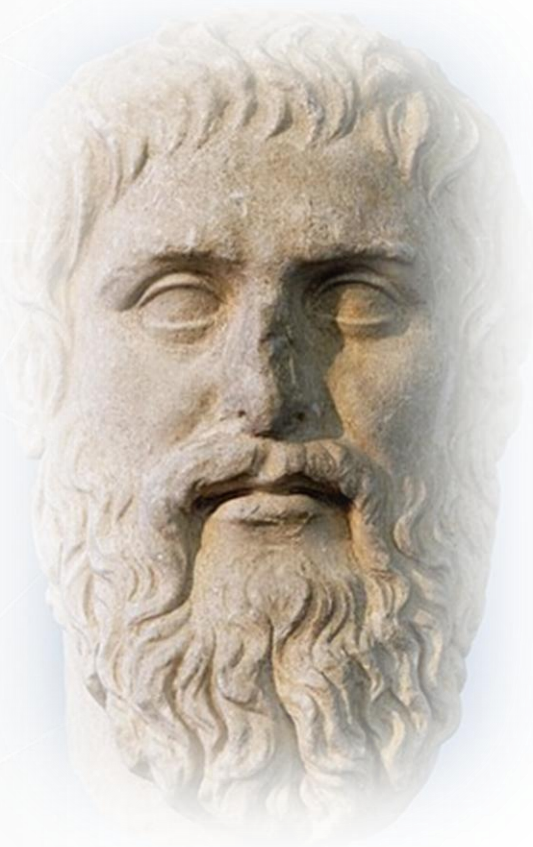
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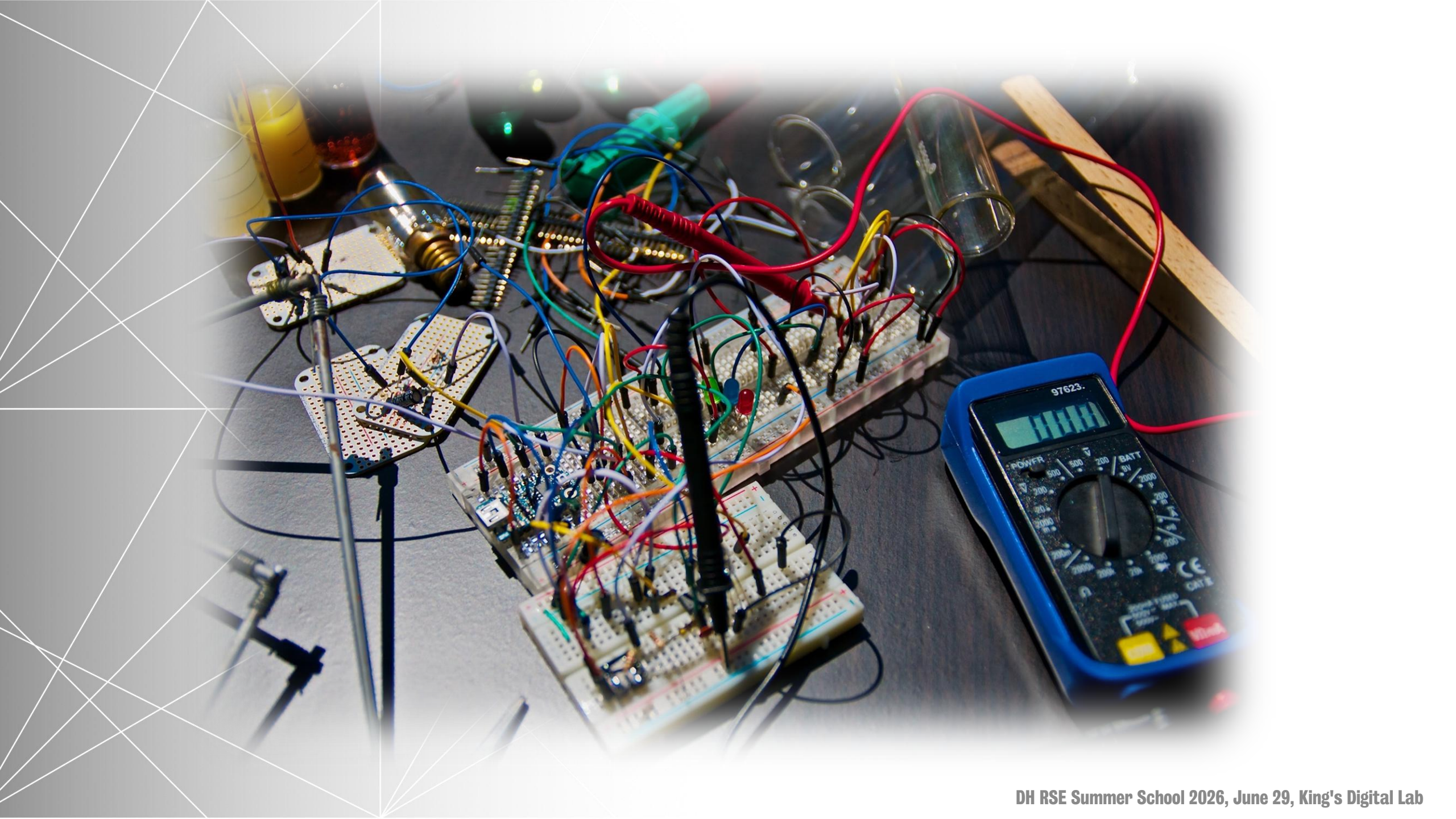


Why are we here?



Why are *you* here?







Minimum Viable Product (MVP)

*“An output which meets the essential needs of the research project and thus doesn’t **compromise** the **rationale** of commencing the project in the first place!”*

Neil Jakeman, DH RSE Summer School 2026 29/06/2026 (approx. 10:45am)





The Iceberg Illusion

Success is an iceberg

SUCCESS!

WHAT PEOPLE SEE

Persistence



Failure



Sacrifice



Disappointment



WHAT PEOPLE DON'T SEE

Dedication



Hard work



Good habits

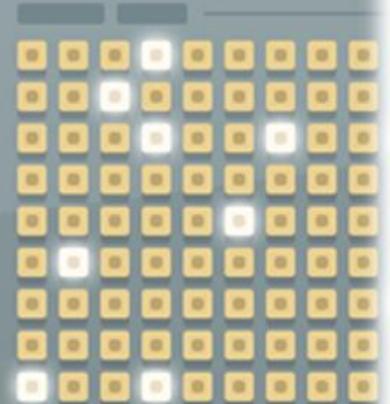
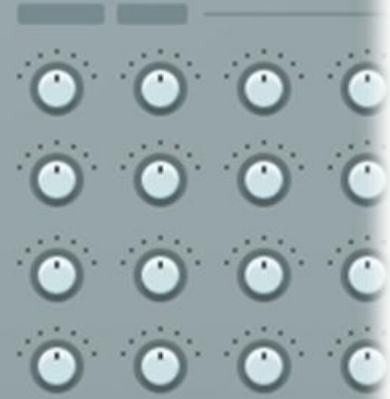


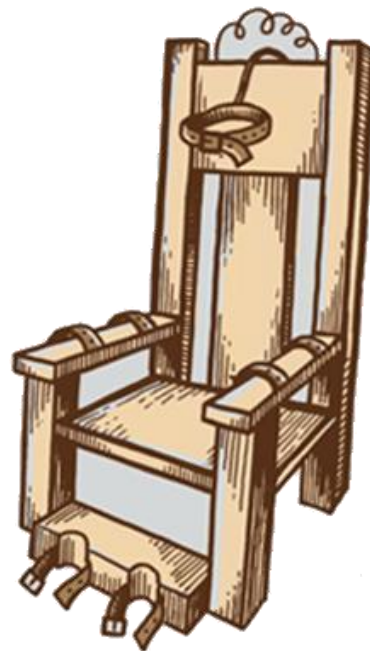
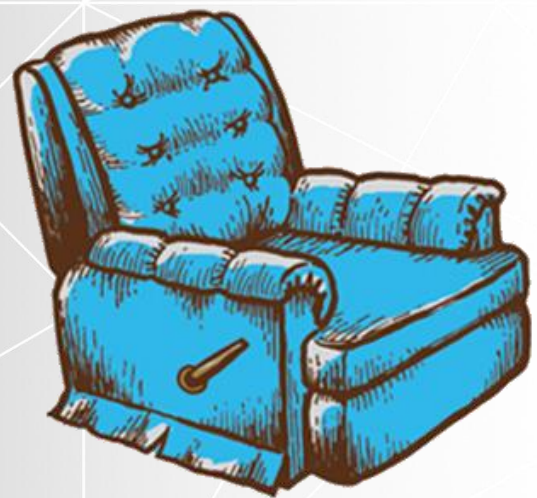
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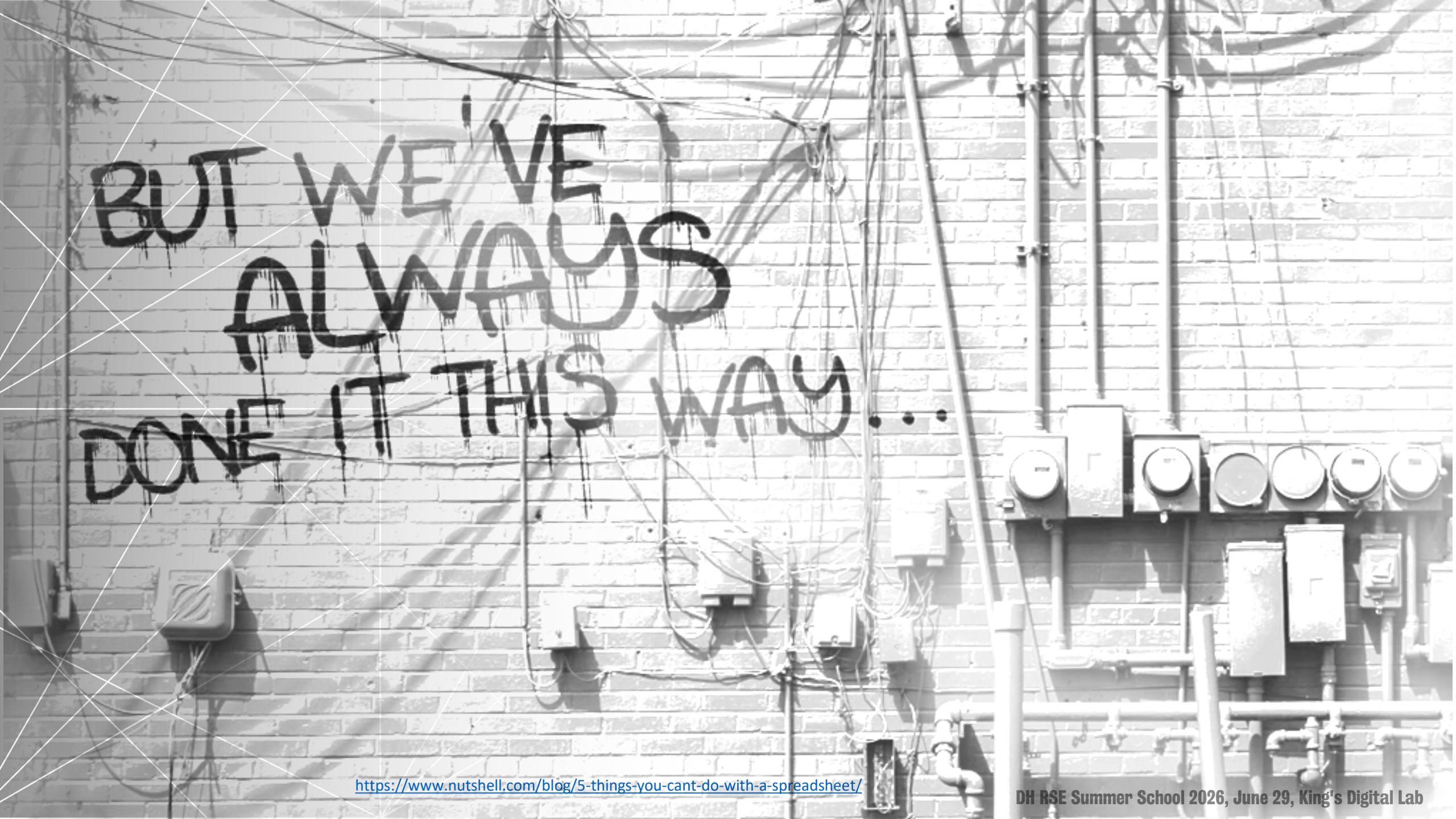
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Select filter to filter by







BUT WE'VE
ALWAYS
DONE IT THIS WAY...

<https://www.nutshell.com/blog/5-things-you-cant-do-with-a-spreadsheet/>

DH RSE Summer School 2026, June 29, King's Digital Lab

The SDLC evolved to:

- Ensure a rigorous project **feasibility** evaluation is undertaken
- Provide a clear product quote in which the **prioritised** requirements of the project are listed
- Manage the development period to ensure that **changing priorities** can be accommodated
- Validate the requirements defined both at the outset of the project, and to **iteratively** review requirements over the development period
- Make sure that a functioning **Minimum Viable Product** is delivered
- Provide a **Service Level Agreement** that defines how the output will be sustained for a defined period
- Provides options for graceful retirement and archiving of a digital resource
- Define RSE skills and roles in a way that can be used to promote a **career path and staff continuity and development**

The Marvellous
Research Software
Development Team!

SDLC Roles in DH RSE

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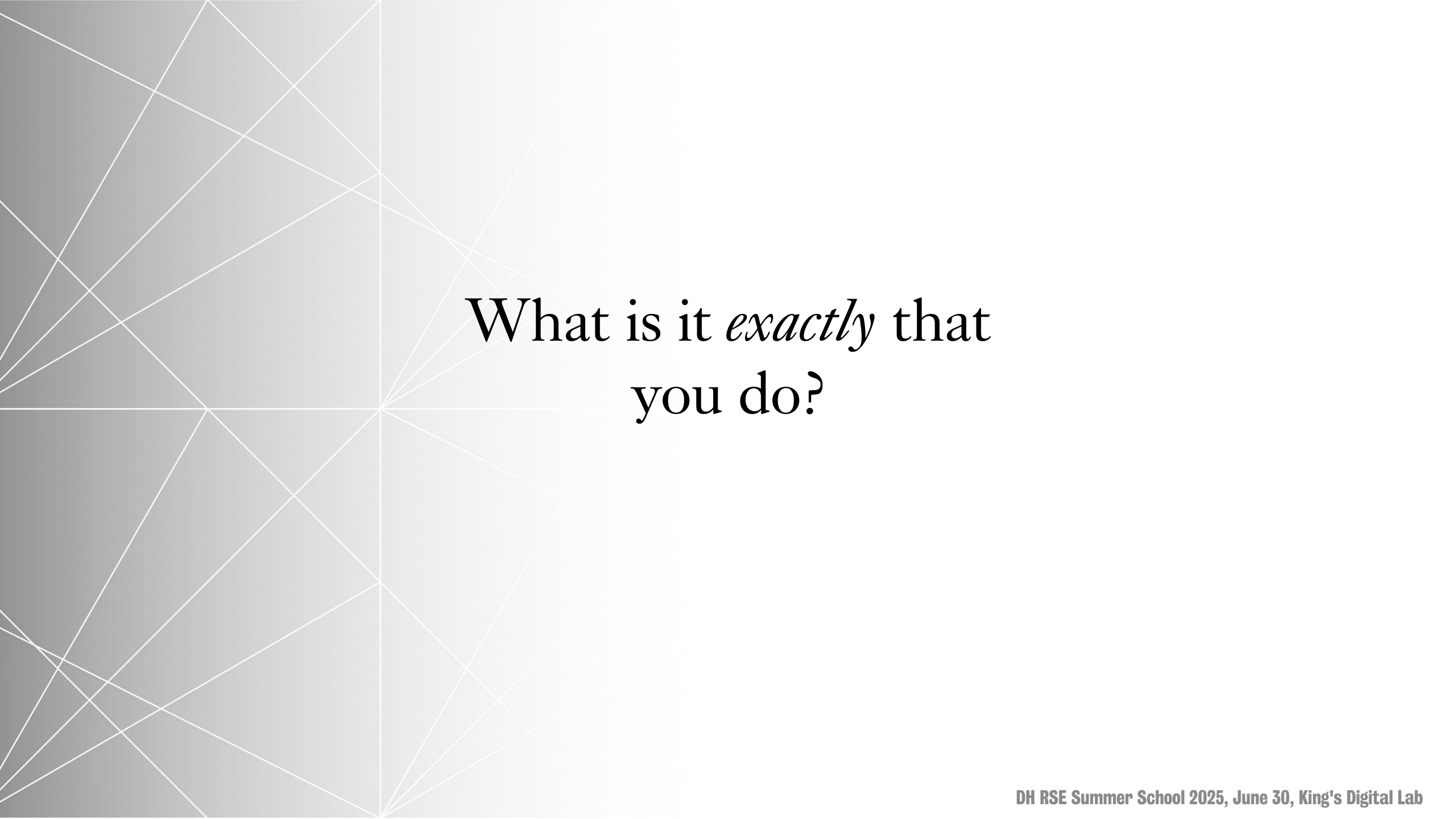


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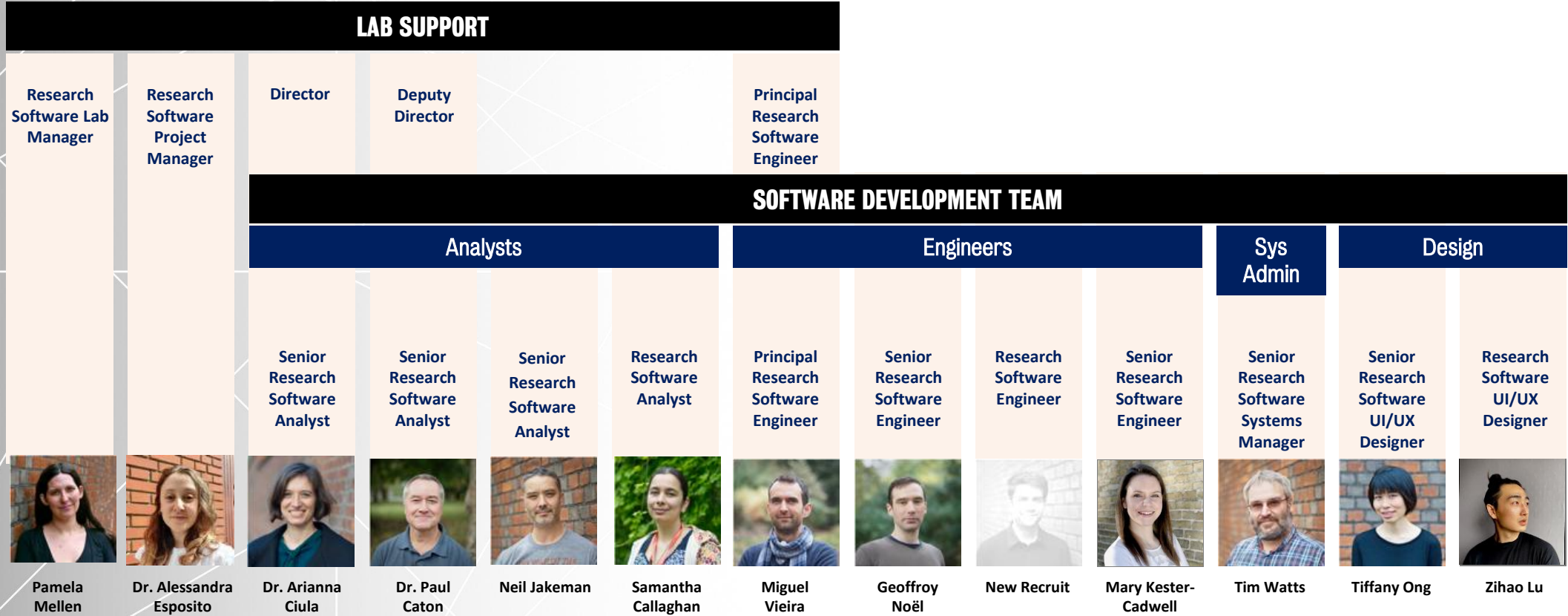
SDLC Roles in DH

What is it that you do
again?



What is it *exactly* that
you do?

Team Structure



What is it *exactly* that you do?

DREAM-ENABLER
EXPECTATION-CRUSHER
REALITY-WARDEN



- ♦ **CODE-WEAVER**
- ♦ **FEATURE-FORGER**
- ♦ **BUG-BANISHER**



- ♦ FLOW-SHAPER
- ♦ SCREEN-SCULPTOR
- ♦ USER-MINDER



- ♦ <CHAOS-HERDERS
- ♦ DEADLINE-DANKERS
- ♦ SCOPE-WATCHERS



- ♦ SERVER-WHISPERER
- ♦ NETWORK-TAMER
- ♦ UPTIME-GUARDIAN



- ♦ NEED-FINDER
- ♦ PATH-SEEKER
- ♦ GEEK-TRANSLATOR
- ♦ SALES-PREACHER
- ♦ FUNDING-GNOME
- ♦ STRAND-CONNECTOR



Role	Core Responsibilities	Key Skills	Overlap With
 Analyst	Understands research questions, gathers requirements, translates needs into specs	Interviewing, requirements elicitation, critical reading, logical structuring	Project Manager (planning), Developer (spec writing), Designer (UX goals)
 Developer	Writes, tests, and maintains software	Programming (Python, JS, etc.), Git, debugging, documentation	Analyst (specs), Designer (UX implementation), Systems Manager (deployment)
 Designer	Designs user-facing elements: interfaces, workflows, user experience	Wireframing, accessibility, user testing, information architecture	Developer (frontend code), Analyst (user needs), PM (prioritisation)
 Systems Manager	Manages infrastructure: servers, deployments, security,	DevOps, containerisation (Docker), backups, security, uptime monitoring	Developer (build pipelines), PM (release scheduling)
 Project Manager	Plans timelines, manages communication & delivery	Agile/DSDM, task breakdown, facilitation, risk management	Analyst (scope), Developer (estimates), all roles (coordination)

Initial Contact

Provide simple channels for approach that require essential information regarding the research, the project goals, the funding, the expectations, the timeframes etc.

Topical?
Defines
new
standard?

Internal Assessment

Decide as a team how and if the project meets strategic and intellectual goals. Discuss chances of success, resources needed etc. Should the project progress beyond this stage?

Decision point

Requirements Assessment

Meet with the research team to discuss the project further, completing Terms of Reference and Feasibility documents and associated processes. Ensure research team understand the development protocols of the RSE team

more time
for
testing
needed

about
archiving?

Funding Application

Help to ensure the application is of an excellent standard, possibly providing contributions to the relevant sections of the application, underwriting the technical approach

Decision point

After analysis is there a satisfactory strategy to meet the project requirements?

Please prep
Data Mgmt
Plan
ASAP

Kick-off

If funding is successful, schedule a kick-off meeting and plan out the first increments of development with the research team.

Release

The project does not go live until a Service Level Agreement is in place with the appropriate stakeholder, which will detail respective responsibilities and commitments in the post project period.

Post Project

The project is maintained under the terms of the SLA until it expires, at which point the continued maintenance of the project may become the subject of a comprehensive Statement of Work prior to any new agreement being undertaken.

Evolutionary Development

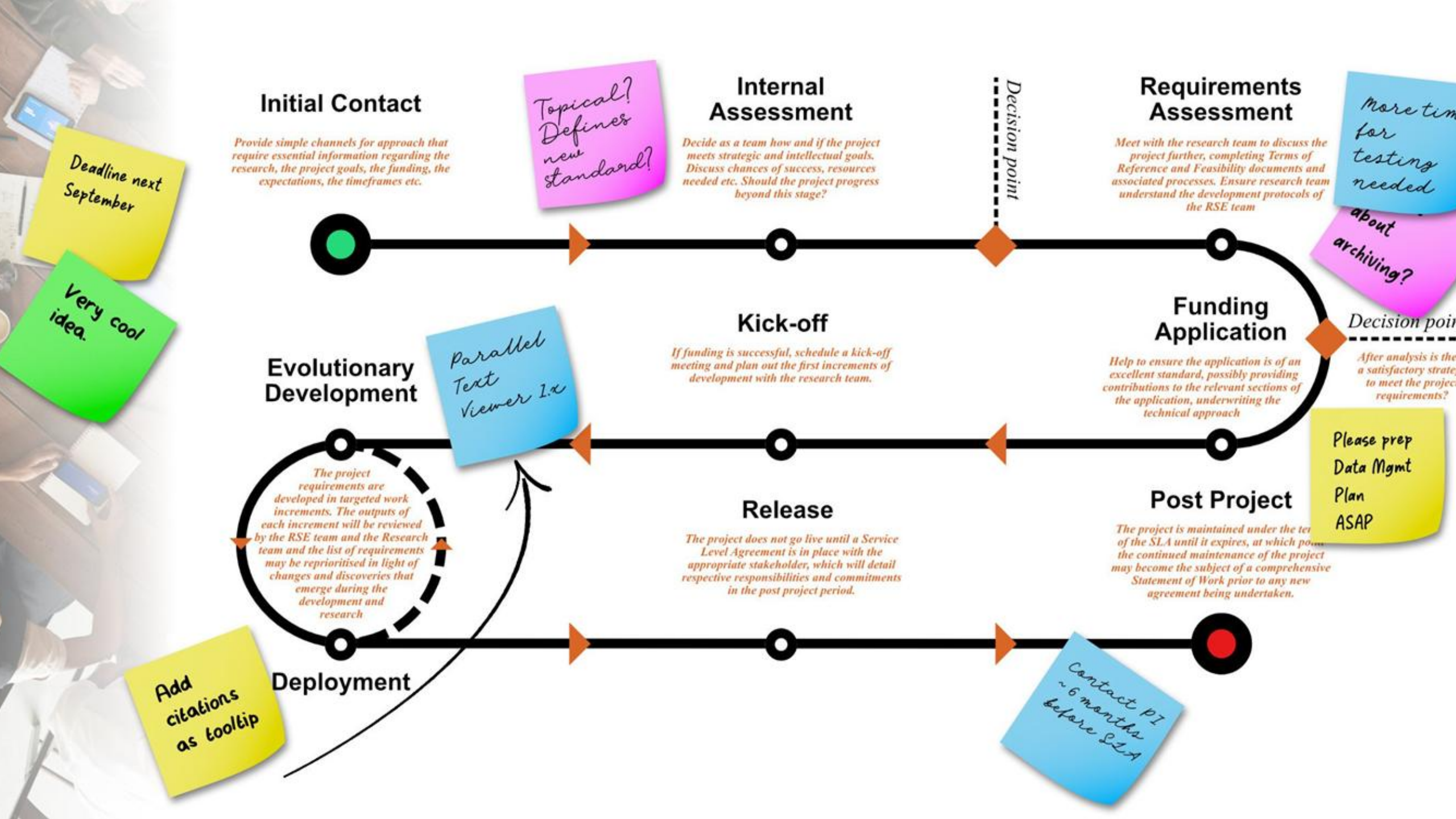
The project requirements are developed in targeted work increments. The outputs of each increment will be reviewed by the RSE team and the Research team and the list of requirements may be reprioritised in light of changes and discoveries that emerge during the development and research

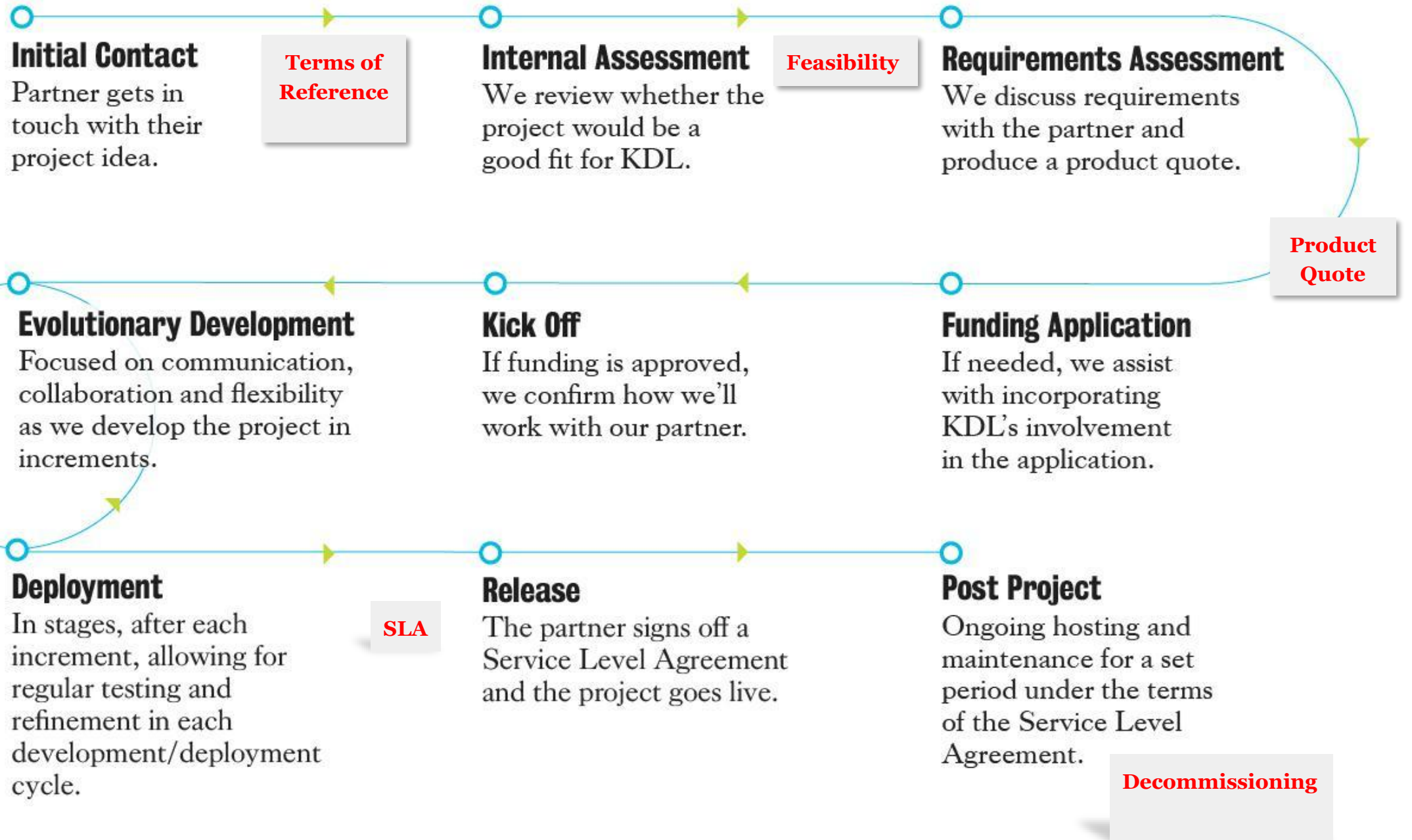
Parallel
Text
Viewer 1.x

Deployment

Add
citations
as tooltip

Contact PI
~6 months
before SLA





Home

James Smithies edited this page on Apr 21 · ·

A Software Development Software Engineering

A toolkit for documentation
a research setting.

This repository was established by King's
piece of work, enriched with community

The documentation found here represents
has been pragmatically adapted to suit
Engineering (RSE).

How to use this resource

Document guidance and templates have
most desktop word processing software
used offline.

1. Getting started

The core component of this toolkit is the
the SDLC. In addition to the document
in the document templates and what
by KDL's experiences in many projects

A quick note on naming conventions

The document names are prefixed with
project setting. The letter is followed by
accompanying guidance document. *NB*
will be required for all projects.

Abbreviations used

Single letters are frequently used as
Developer ID= Software Developer ID

- Document templates

- A1: Terms of Reference
- B1: Project Approach Questionnaire
- F1: Feasibility
- I1: Product Quote
- J1: Statement of Work
- N1: Web Hosting and Infrastructure Service Level Agreement

- Document guidance

- A2: Terms of Reference guidance
- B2: Project Approach Questionnaire guidance
- F2: Feasibility guidance
- I2: Product Quote

► Pages 42

- Home
- Document templates
 - A1: Terms of Reference
 - B1: Project Approach Questionnaire
 - F1: Feasibility
 - I1: Product Quote
 - J1: Statement of Work
 - N1: Web Hosting and Infrastructure Service Level Agreement (SLA)
 - Q1: Decommissioning Authorisation
 - L1: Project Review Record
 - E1: Costings table
- Document guidance
 - A2: Terms of Reference guidance
 - B2: Project Approach Questionnaire guidance
 - F2: Feasibility guidance
 - I2: Product Quote guidance
 - J2: Statement of Work guidance
 - Data Management Plan guidance and AHRC template
 - L2: Project Review Record guidance
 - N2: Web Hosting and Infrastructure Service Level Agreement
 - Q2: Decommissioning Authorisation guidance

• Modelling Methods - In

Typical MoSCoW prioritisation

M = must have; S = should have; C = could have; W = won't have (this time)

Priority	Requirement	Paul Caton (KDL), <i>SHARC project</i> , KDL Product Quote
M	Taxonomic data model for Shakespeare-related items in the Royal Collections and Royal Archives	
M	Metadata schema that facilitates multiple associations among records	
M	Site that can store, search across, and display a set of digital objects representing those items (likely to be approx 2500 objects)	
M	Site that can store, search across, and display a set of metadata records associated with the digital objects	
S	Admin interface that allows direct metadata record creation on site	
S	Map functionality showing location of items by royal residences	
S	Timelines placing items in historical context	
C	Integration of 3D visualizations of key rooms at Windsor Castle (creation of 3D images would be by 3rd party)	
W	Public interaction with/contributions to site	

Typical MoSCoW ratios

In scope
for this timeframe

(Project / Increment / Timebox)

Must Have



Typically
no more
than
60% effort

Should Have



Could Have



Typically
around
20% effort

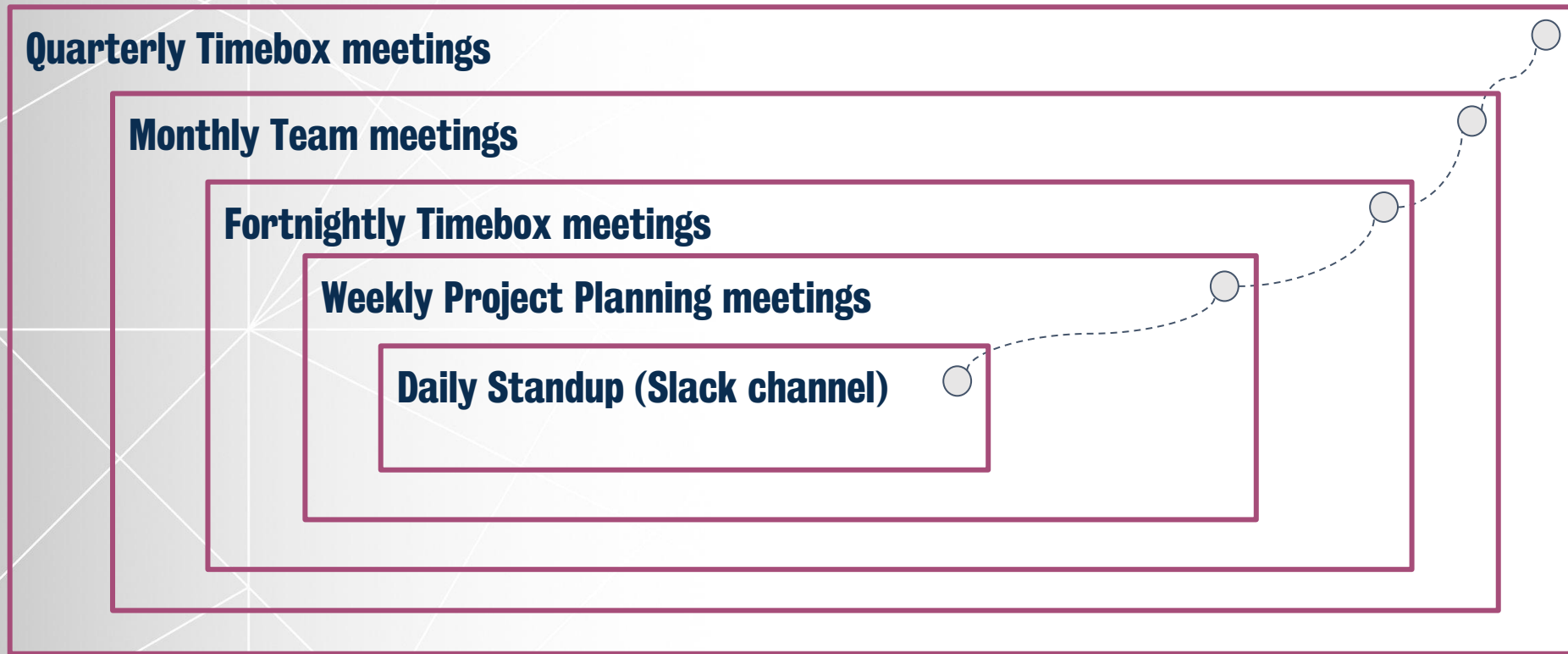
Out of scope
for this timeframe

Won't Have this time



MoScoW technique >> <https://www.agilebusiness.org/content/moscow-prioritisation>

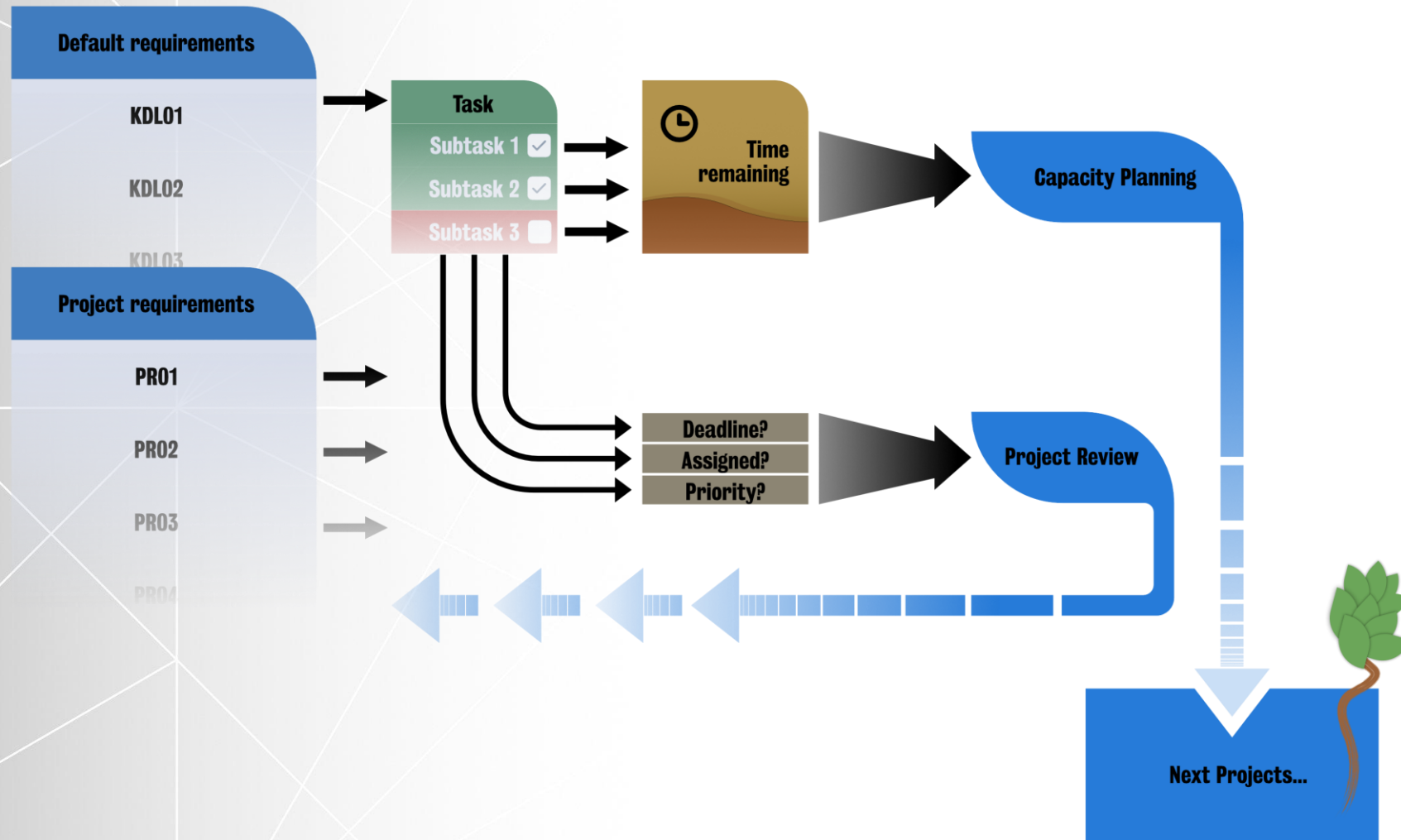
Project Management & Communication



PRE-PROJECT	FEASIBILITY	FOUNDATIONS	EVOLUTIONARY DEVELOPMENT	DEPLOYMENT	POST-PROJECT
Understand the research questions and context Discuss with Project Partners Produce a “Term of Reference” document GO ►	Outline solution architecture development and delivery plan Prioritise Requirements List Produce a summary for the final go/no-go decision Quote GO ►	High-level schedule of work increments	Timebox planning Daily stand-up meetings Project review record Evolving solution document(s) and artefacts	Assemble Review Deploy	Assess if the expected benefits have been met



Requirements, Tasks and Capacity



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RESEARCH CASE

Questions and prompts

- What is the “*elevator pitch*” for the project?

If the PI only had 30 seconds to communicate the idea, how would they do that?

- What is it trying to achieve at its core?

How will the research domain be enriched through this work?

- How do digital methods support these aims?

Is there a risk of over-engineering a technical solution to meet a simple need?

- What are the tangible digital outputs planned for?

How should audiences engage with the completed research?

What will be the legacy of this project and how to we plan to protect it?

RSE Stakeholder Interview Simulator

Scenario

Social Dynamics of War

Difficulty

Realistic

Provider

KCL ARC-AI

Model

arc:lite

DR JAKE NEILMAN

Good morning. I'm Dr Jake Neilman — I'm a professor of war studies here at the university. I've been told your team can help me think through the technical side of a research project I'm developing. I should say upfront, digital methods aren't really my area, but I'm told you're the people to talk to. Where shall we start?

Type your message...

Send

Reset

Session name...

Save

Load session...

MoSCoW Requirements

Export

Clear All

Token budget: 0 / 15

MUST HAVE

0%

No requirements yet

Add to Must Have...

SHOULD HAVE

0%

No requirements yet

Add to Should Have...

COULD HAVE

0%

No requirements yet

Add to Could Have...

WON'T HAVE

0%

No requirements yet

Add to Won't Have...

RSE Stakeholder Interview Simulator

0 min

Scenario

Social Dynamics of War

Difficulty

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arclite

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YOU

What sort of data are you dealing with?

DR JAKE NEILMAN

The core of my research relies on **handwritten army personnel records** from the Second World War. These are essentially standardised forms, but because they were filled in by thousands of different clerks, there is an enormous amount of variation in the handwriting and the way things are recorded.

I am particularly interested in how we might use automated approaches to extract information from these forms at scale. I want to move beyond just looking at individual stories to seeing the broader patterns across the whole population.

Type your message...

Send

Reset

Session name...

Save

Load session...

MoSCoW Requirements

Export

Clear All

Token budget: 0 / 15

MUST HAVE

0%

No requirements yet

Add to Must Have...

SHOULD HAVE

0%

No requirements yet

Add to Should Have...

COULD HAVE

0%

No requirements yet

Add to Could Have...

WON'T HAVE

0%

No requirements yet

Add to Won't Have...

RSE Stakeholder Interview Simulator

1 min

Scenario

Social Dynamics of War

Difficulty

Realistic

Provider

KCL ARC-AI

Model

arc:lite

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Type your message...

Send

Reset

Session name...

Save

Load session...

MoSCoW Requirements

Export

Clear All

Token budget: 1 / 15

MUST HAVE

100%

▼ × Data Extraction st Scale

1t

Confidentiality must be considered

Add a note or rationale...

Add to Must Have...

SHOULD HAVE

0%

No requirements yet

Add to Should Have...

COULD HAVE

0%

No requirements yet

Add to Could Have...

WON'T HAVE

0%

No requirements yet

Add to Won't Have...

RESEARCH CASE

Questions and prompts

- What distinguishes this research as particularly ambitious or innovative?
- Does the project align with current internal goals and strategies?
- Does the project align with internal research interests?
- Does the project offer an opportunity to engage with new technology and approaches?
- Does the project offer an opportunity to develop a mutually beneficial strategic relationship with an interesting partner?
- Does the project align with your institutional values and ethical considerations?
- Have the RSE team had **previous experience** of working with this project team or members of the team?
- Will the project team fully **engage with the SDLC** processes?

BACKGROUND CONTEXT

Questions and prompts

- What **previous work** (if any) is being built upon?
- Has previous work in this area failed to produce quality research? If so, why?
- Is there an **existing resource** upon which this new work depends?
- Does the RSE team have access to **the necessary expertise** to fulfil the project requirements?
- How amenable to digitisation or to computational processing are the data sources being proposed?
- Are **there sample data** to access and assess?
- Where are the datasets hosted if they already exist or where is the project team expecting to host them, and for how long?
- How amenable to archiving is the project being proposed?
- Are the ambitions of the team realistic?
- Does the project present **ethical concerns**?

SOLUTION ARCHITECTURE

Questions and prompts

- Does the proposed technical solution take into account data standards?
- Does it, or will it ultimately, adhere to [FAIR principles](#)?
- Is it clear that the RSE team has sufficient expertise to implement the solution?
- Is the solution scalable? Does it need to be?
- Which RSE roles will be performing which tasks?

DEVELOPMENT APPROACH

Questions and prompts

- Where will the evolving solution be hosted?
- What backup strategies are in place?
- Which channels and protocols should the research team use to seek or provide feedback?
- How and when will review meetings take place?
- When will *change freeze* be implemented?

- **What is our stakeholder trying to achieve?**
- **What else would it be helpful to know?**
- **What technical considerations can we infer?**
- **What initial creative ideas does this prompt?**
- **What sort of requirements (MoSCoW?) can we draw from this?**

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